PORTFOLIO PROJECT · HEALTHCARE CLAIMS AUTOMATION

FWA Claims Rule Engine

A rule engine that screens health-insurance claim exports for fraud, waste and abuse. It runs 32 rules over every claim and service line, writes a reason next to each finding, and adapts to whatever layout the claims export arrives in.

This example is built for the Saudi market, using NPHIES service codes (SBS), ICD-10, the SFDA drug register and the NPHIES GTIN list. The rules are not tied to them: the code lists can be swapped for other standards, such as CPT, HCPCS, ICD-10-CM or NDC drug codes.

32 validation rules | **5** claim-level | **27** service-line level

Any export layout or file size

Flowchart for every rule

The problem

Medical claim audit is mostly manual. Reviewers sample claims and look for the same patterns again and again: the same service billed twice, a diagnosis that cannot apply to the patient's age or sex, one claim copied across many members, drugs that duplicate each other, a visit billed after the policy expired. Each check is simple; doing all of them on every line of a large export is not.

This project turns those checks into explicit, documented rules that run on the whole file in one pass, against the coding standards providers bill with: NPHIES service codes (SBS), ICD-10 diagnoses, the SFDA drug register and the NPHIES GTIN list.

How it works

STEP 1

Map the export

A column converter matches the file's headers to a 60-column claims template. Headers are mapped once in the app and remembered; older export spellings are recognised automatically. Service-type wording (CONS, MEDS, LBRD…) maps to seven types.

STEP 2

Load the standards

SBS, ICD-10, the SFDA drug list and the NPHIES GTIN list sit next to the app and are reused every run. Uploading a newer version replaces the stored copy.

STEP 3

Run the rules

Unpaid lines are set aside, then the claim-level rules and the service-line rules run in a fixed order, so a rule that depends on another always sees its result.

STEP 4

Deliver the findings

An Excel workbook with a reason written next to every flagged line, in the layout the reviewer chose before the run, ready to hand to the audit team.

STEP 5

Track it on the dashboard

Alongside the workbook, an interactive dashboard turns the findings into money and trends: billed against rejected amount, rejections by rule and provider, and how both move month by month.

Why the dashboard matters

A findings list says what is wrong; the dashboard says where to act first. It shows how much of what was paid the rules would reject, which providers and which rules drive that amount, and whether a pattern is growing or fading over time. That turns screening into something a manager can prioritise and report on: rejection rate, savings by rule, and providers to audit or educate.

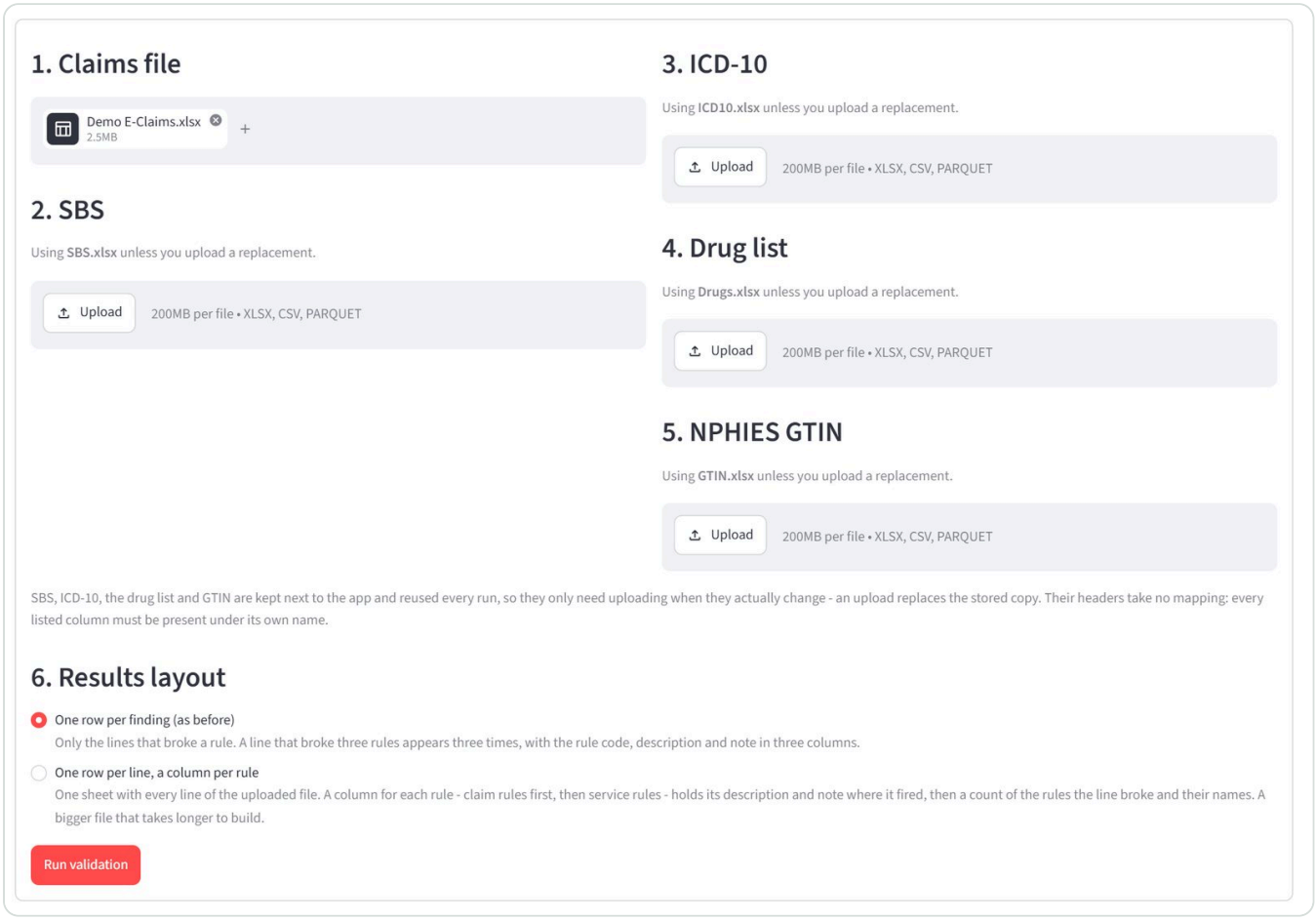
- **Headline cards** — total paid and rejected amount, with the rejection rate. A line that breaks several rules is counted once, so the rejected amount is never inflated.
- **Billed and rejected over time** — one line per provider, by month, quarter or year, to spot a provider whose billing or rejections jump.
- **Rejected amount by rule** — which rules carry the money, at claim or service level; any rule can be hidden so the others rescale into view.
- **Provider behaviour** — each provider's share of unlisted codes against its limit, and the services a provider or physician attaches to most of their claims.
- **Filters** — provider, period (year, quarter, month or a custom range) and service type, applied across the page.

See it in action

> Screens from the working app, run on a synthetic claims file built for this portfolio. Every patient, physician, provider and claim in them is invented; only the medical and drug codes are real public codes.

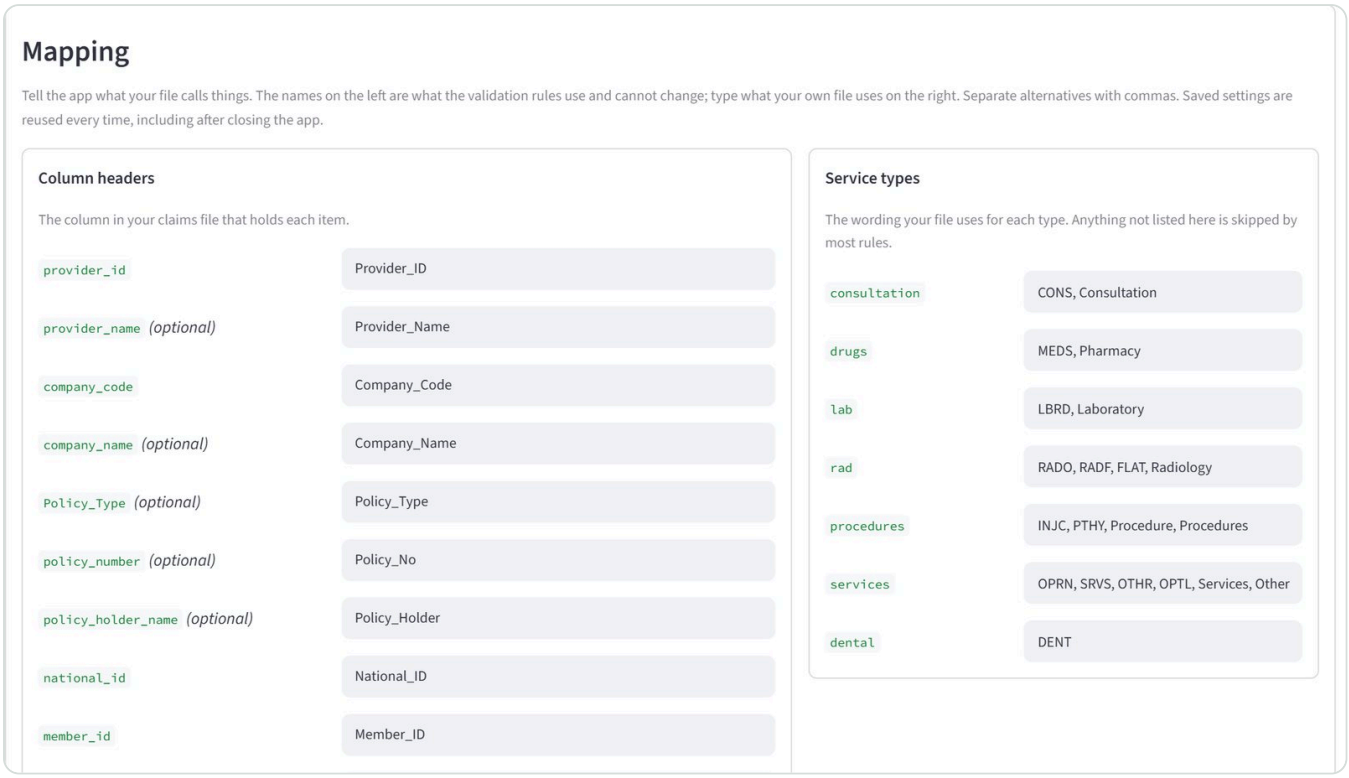
Run screen

Upload the claims file; the four coding standards are stored and reused. The results layout is chosen before the run.



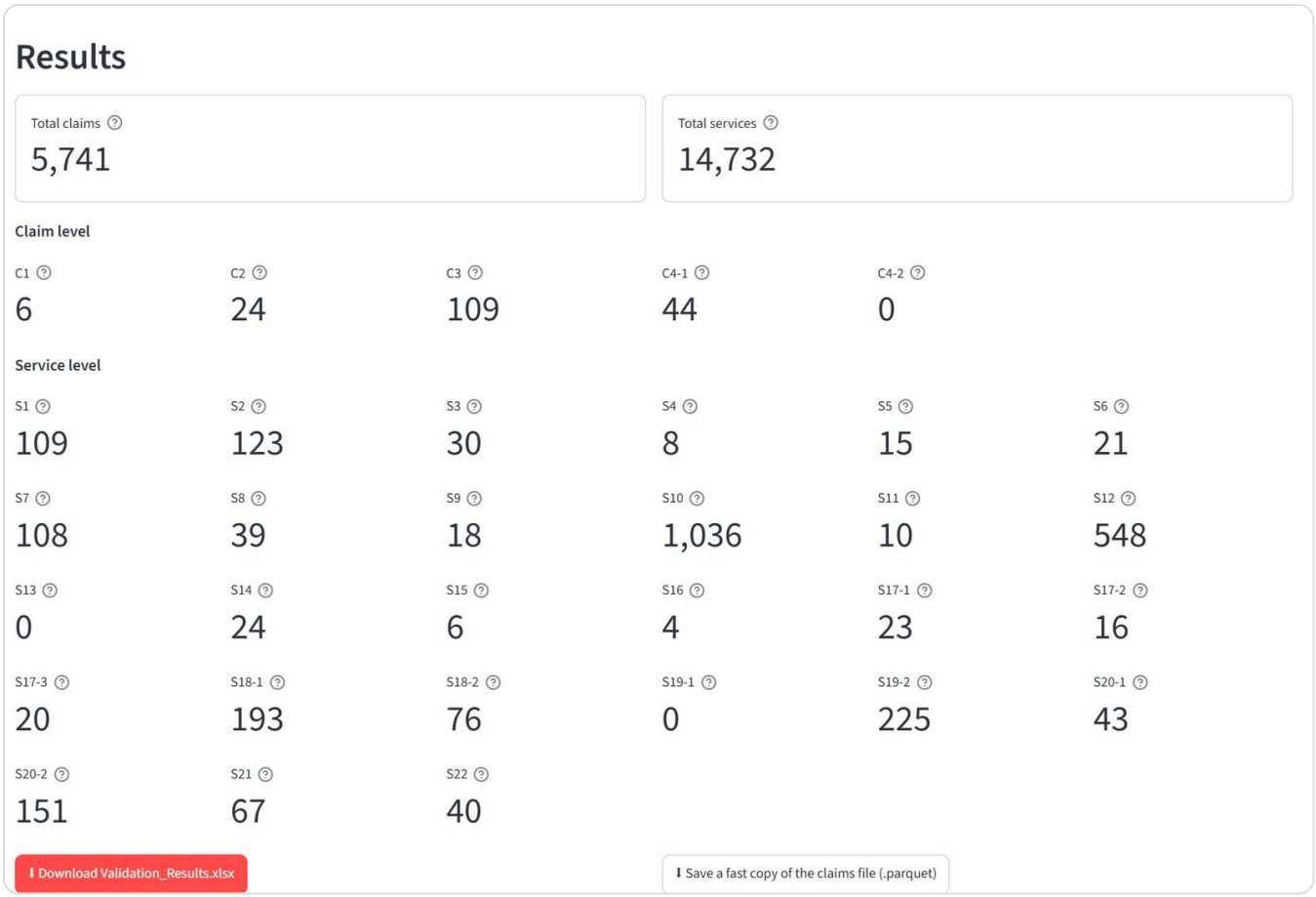
Column converter

Each template field is mapped once to the file's own header, and the file's service-type wording to seven types. Saved and reused on every run.



Results

What the run covered - total claims (distinct claim IDs) and total service lines - then a count for every rule, claim level and service level, and the workbook download.

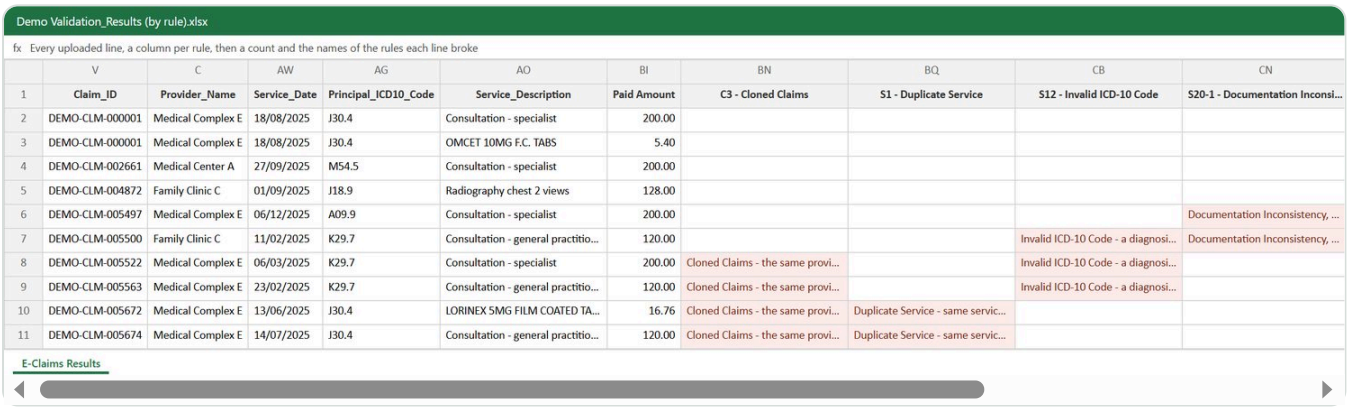


What the reviewer gets

Two download layouts. **One row per finding** lists each broken rule with its code, description and note. **One row per line** keeps every line of the uploaded file and adds a column for every rule, claim rules first, then a count and the names of every rule the line broke:

Results workbook: one row per line

A slice of the real workbook from the demo run: 14,732 lines across 97 columns. The letters above the headers are the columns' positions in the full sheet; rule cells hold the rule's description and its note.



Demo Validation_Results (by rule).xlsx

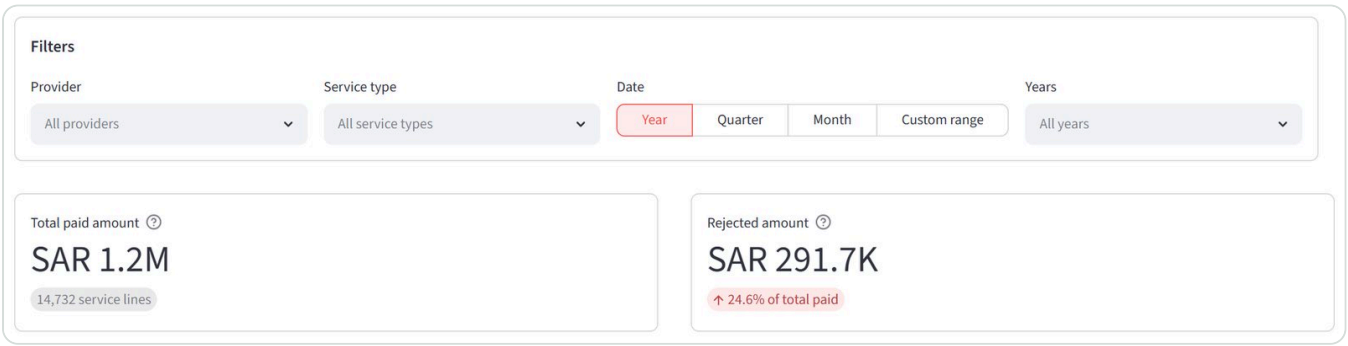
fx Every uploaded line, a column per rule, then a count and the names of the rules each line broke

	V	C	AW	AG	AO	BI	BN	BQ	CB	CN
1	Claim_ID	Provider_Name	Service_Date	Principal_ICD10_Code	Service_Description	Paid Amount	C3 - Cloned Claims	S1 - Duplicate Service	S12 - Invalid ICD-10 Code	S20-1 - Documentation Inconsi...
2	DEMO-CLM-000001	Medical Complex E	18/08/2025	J30.4	Consultation - specialist	200.00				
3	DEMO-CLM-000001	Medical Complex E	18/08/2025	J30.4	OMCET 10MG F.C. TABS	5.40				
4	DEMO-CLM-002661	Medical Center A	27/09/2025	M54.5	Consultation - specialist	200.00				
5	DEMO-CLM-004872	Family Clinic C	01/09/2025	J18.9	Radiography chest 2 views	128.00				
6	DEMO-CLM-005497	Medical Complex E	06/12/2025	A09.9	Consultation - specialist	200.00				Documentation Inconsistency, ...
7	DEMO-CLM-005500	Family Clinic C	11/02/2025	K29.7	Consultation - general practitio...	120.00			Invalid ICD-10 Code - a diagnosi...	Documentation Inconsistency, ...
8	DEMO-CLM-005522	Medical Complex E	06/03/2025	K29.7	Consultation - specialist	200.00	Cloned Claims - the same provi...		Invalid ICD-10 Code - a diagnosi...	
9	DEMO-CLM-005563	Medical Complex E	23/02/2025	K29.7	Consultation - general practitio...	120.00	Cloned Claims - the same provi...		Invalid ICD-10 Code - a diagnosi...	
10	DEMO-CLM-005672	Medical Complex E	13/06/2025	J30.4	LORINEX 5MG FILM COATED TA...	16.76	Cloned Claims - the same provi...	Duplicate Service - same servic...		
11	DEMO-CLM-005674	Medical Complex E	14/07/2025	J30.4	Consultation - general practitio...	120.00	Cloned Claims - the same provi...	Duplicate Service - same servic...		

E-Claims Results

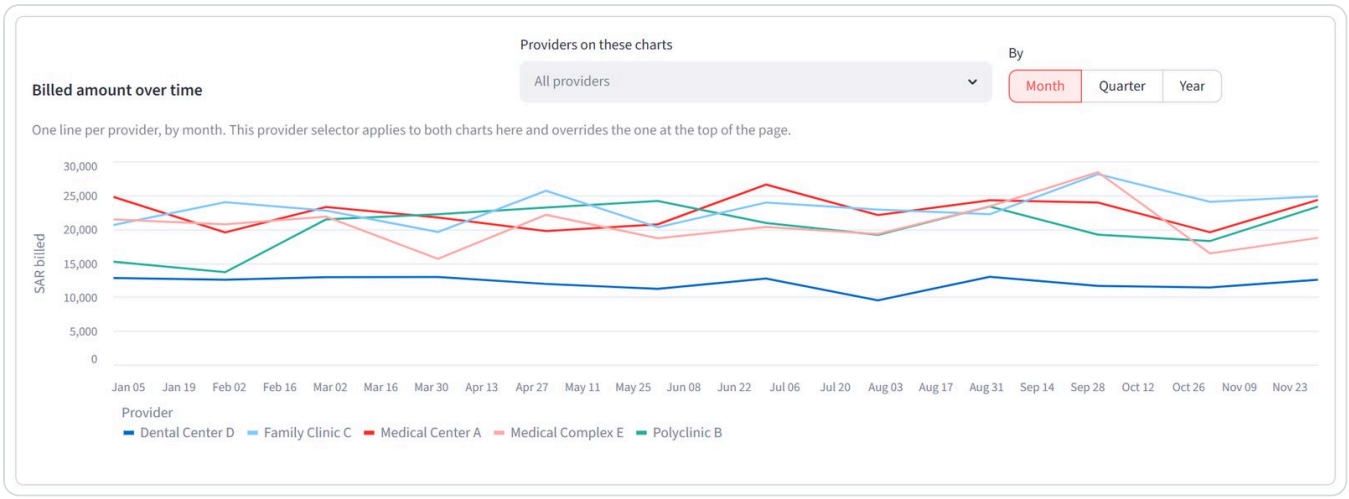
Dashboard: overview

Filters for provider, period and service type; total paid against rejected, with the rejection rate.



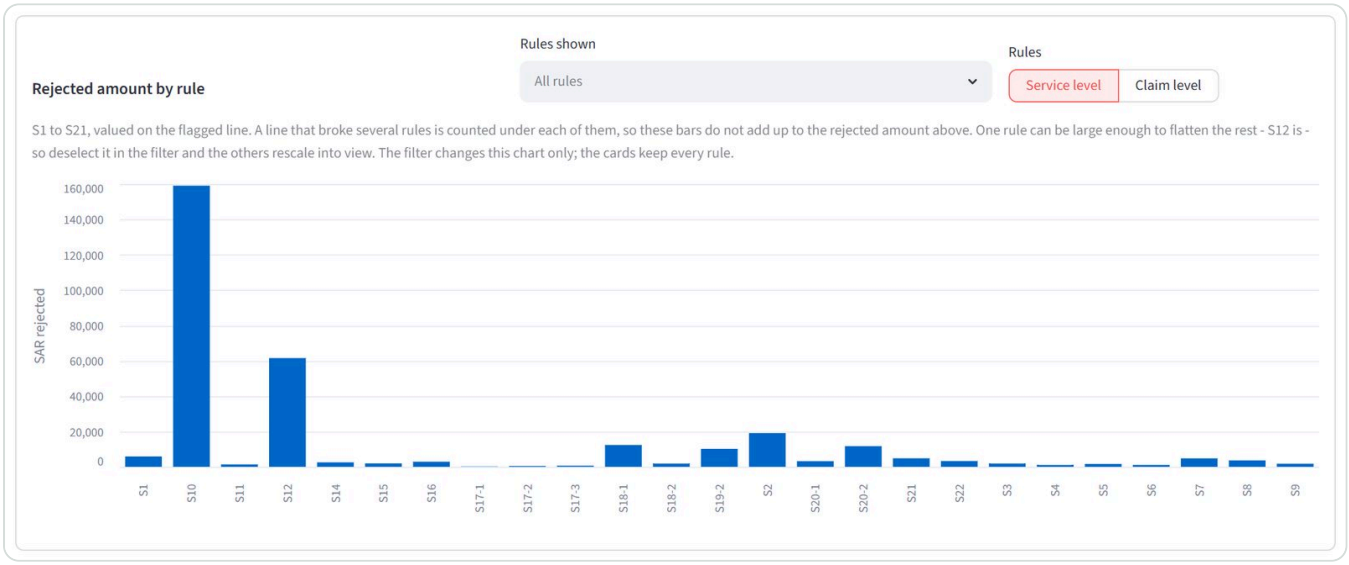
Dashboard: trend by provider

Billed amount per provider by month, quarter or year. A matching chart below it shows the rejected amount.



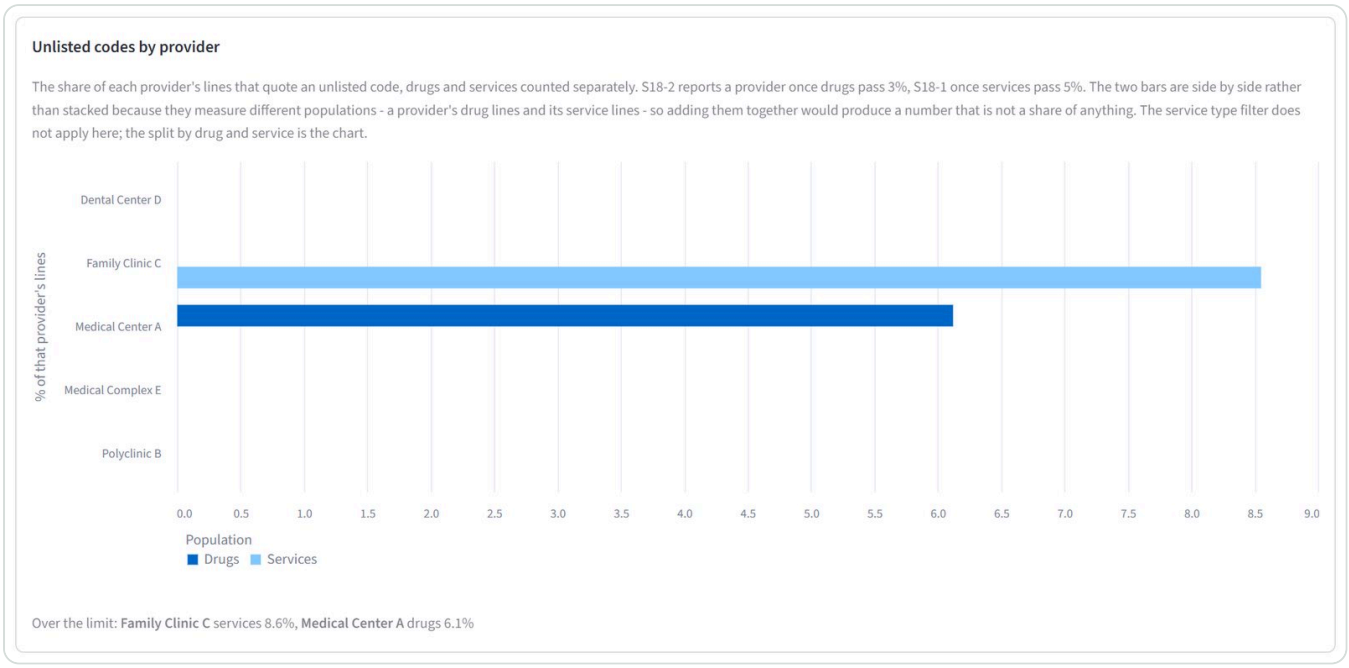
Dashboard: where the money is

Rejected amount by rule, at service or claim level. Any rule can be hidden so the others rescale into view.



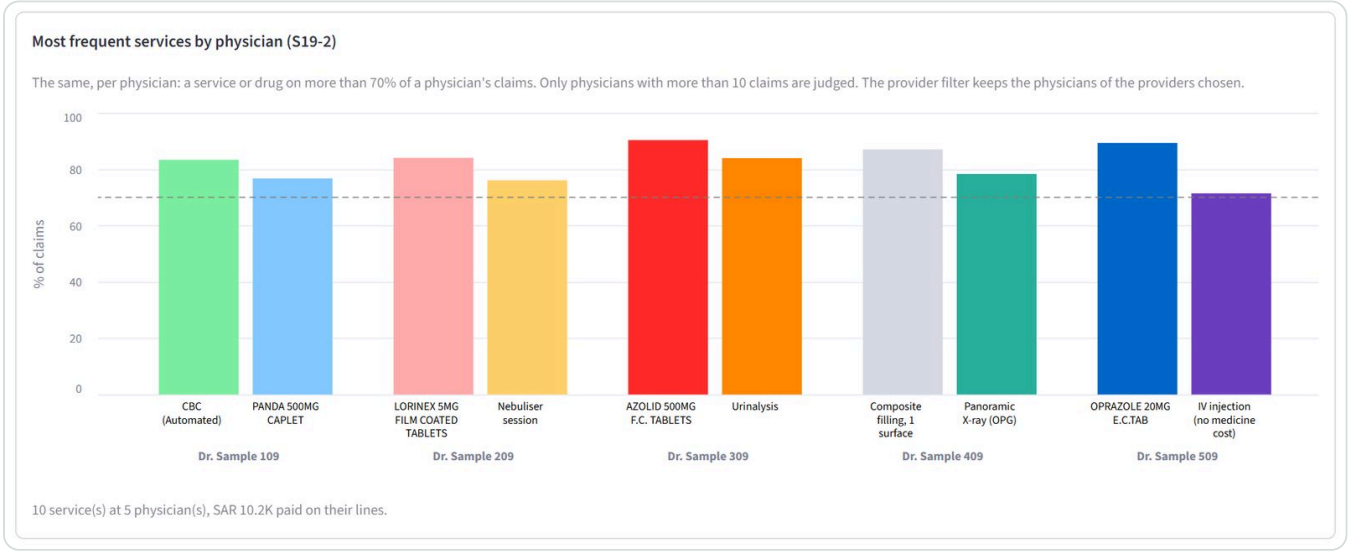
Dashboard: provider behaviour

Each provider's share of unlisted codes, drugs and services separately, against the limits the S18 rules apply.



Dashboard: physician pattern

Physicians attaching the same services or drugs to most of their claims - each doctor's items side by side, every bar above the 70% line of rule S19-2.



The 32 rules

Grouped by what they look for. Each has a flowchart, a reference entry and a note template, so every finding says why it was raised.

See the flowchart for every rule →

Claim level 5 rules

C1	**Duplicate Claim** Two or more claims for the same patient on the same day billing exactly the same set of services.
C2	**Member Treated at Multiple Providers, Same Day** A patient with paid consultation lines at two or more different providers on one day.
C3	**Cloned Claims** The same provider, anywhere in the file, billing an identical set of services, principal diagnoses and chief complaint for different patients.
C4-1	**Physician Daily Capacity Breach (doubtful)** A physician seeing more than 40 different patients in consultation on one day.
C4-2	**Physician Daily Capacity Breach (confirmed)** A physician seeing 60 or more different patients in consultation on one day.

Duplication and frequency 9 rules

S1	**Duplicate Service** The same service billed more than once for the same patient on the same day.
S2	**Repeat Consultation Within 14 Days** The same patient seen again by the same physician 1 to 14 days after the previous visit.
S3	**Service Frequency Breach** A service repeated sooner than the frequency SBS sets for its NPHIES code, counted across providers.
S4	**Excessive Radiology** More than 3 radiology studies for the same patient in a rolling 14 days.

S5 **Exceeds Maximum Daily Limit**

A line billed with a quantity greater than the daily limit SBS sets for its NPHIES code.

S6 **Unreasonable Quantity**

A drug line billed with a quantity above 5.

S7 **Duplicated Service Within 7 Days**

The same service for the same patient at the same provider repeated 1 to 7 days later.

S11 **Excessive Consultation Frequency**

A patient with more than 3 consultation-only claims inside a rolling 30 days.

S22 **Excessive Member Visits**

A patient with more than 5 visit days inside a rolling 30 days, counted across every provider and service type.

Coding validity `7 rules`

S8 **Diagnosis Outside Age Range**

A diagnosis code billed for a patient whose age at the service date is outside the range ICD-10 allows.

S9 **Diagnosis Gender Mismatch**

A diagnosis code ICD-10 strictly restricts to one sex, billed for a patient of the other.

S12 **Invalid ICD-10 Code**

A diagnosis code marked valid = 0 in ICD-10, or absent from the reference altogether.

S13 **Inactive ICD-10 Code**

A diagnosis code ICD-10 had withdrawn by the service date and not reactivated.

S14 **Not Valid Principal Diagnosis**

A primary diagnosis that ICD-10 allows only as an additional diagnosis.

S15 **Service / Gender Mismatch**

A service SBS restricts to one sex, billed for a patient of the other.

S16 **Service / Age Mismatch**

A service billed for a patient outside the age range SBS allows it for.

Pharmacy 3 rules

S17-1 **Therapeutic Duplication, Same Active Ingredient**

Two different drugs given to one patient on one day that share the ATC code to 7 characters (same active ingredient).

S17-2 **Therapeutic Duplication, Same Chemical Subgroup**

Two different drugs given to one patient on one day that share the ATC code to 5 characters (same chemical subgroup).

S17-3 **Therapeutic Duplication, Same Pharmacological Subgroup**

Two different drugs given to one patient on one day that share the ATC code to 4 characters (same pharmacological subgroup).

Provider and physician behaviour 4 rules

S18-1 **More Than 5% Unlisted SBS Codes**

Every unlisted service from a provider whose unlisted share of non-drug lines is 5% or more.

S18-2 **More Than 3% Unlisted Drugs**

Every unlisted drug from a provider whose unlisted share of drug lines is 3% or more.

S19-1 **Service Too Frequent Use, Provider Level**

A service or drug that appears on more than 70% of a provider's claims, consultation excluded.

S19-2 **Service Too Frequent Use, Physician Level**

A service or drug that appears on more than 70% of a physician's claims, consultation excluded.

Documentation and coverage 4 rules

S10 **Isolated Consultation Visit**

A claim billing nothing but consultation, with no test, medicine or procedure that followed.

S20-1 **Phantom Billing**

A paid line on a claim whose chief complaint says the visit was cancelled or the patient not seen.

S20-2 **Missed Chief Complaint**

A paid line on a claim with no chief complaint, or one holding no real word.

S21 **Service Outside Coverage**

A service dated before the policy start date or after the policy end date.

Built to fit any claims data

Any layout

- A 60-column template covers member, policy, claim, diagnosis, service, amount and rejection fields.
- A mapping screen ties each template field to the file's own header; the mapping is saved and reused.
- Fields a file does not have are created blank, so a partial export still runs; rules that need a missing field say so instead of failing.
- Results are written back under the file's own header names.

Any size

- No fixed line limit: rules work on whole columns at once rather than row by row.
- A results sheet that passes Excel's 1,048,575-row limit continues on a second sheet automatically.
- A fast copy of the upload (Parquet) makes repeat runs on the same file load in seconds.
- Reference tables can be swapped for newer releases without touching the rules.

How the rules were set and proven

Thresholds from the data

Limits were measured before they were chosen. For example, the excessive-visits rule counts each patient's visit days in a rolling 30 days; visit counts are heavily skewed, so the limit came from Tukey's outlier fences on that distribution rather than a guess.

Tested at the boundary

Every rule has tests built at its exact limit: at the threshold it stays quiet, one past it fires, the wrong service type never fires, and a missing reference leaves it silent with a warning instead of an error.

Checked against a recount

On a full file, each new or changed rule is recounted by a separately written method and compared line by line, and every other rule is confirmed unchanged before the change is kept.

Documented for review

A rules reference workbook, a flowchart per rule, and a shared review page where stakeholders comment on a rule after reviewing its results.

My role and tools

I designed every rule, chose and tested the thresholds against real-world claim patterns, specified how findings are reported, and validated the results. The engine is implemented in Python (pandas) with a Streamlit app and dashboard, built with AI-assisted development.

Python · pandas · Streamlit · Altair · Excel · NPHIES SBS · ICD-10 · SFDA drug register · GTIN

A personal portfolio project. No claims data, patient information or provider details are shown; the example table is fictitious.